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Transfer Suffers NEVER!

We present an investigation of the costs of the airlines and develop a model to keep their costs as low as possible. Our model is divided into three parts:

Pre-requisite: We locate the airport in a grid chart in order to ease the difficulty in locating escorts and the route of the escort. According to the average passenger flow of the airport at each gate, we obtain the optimal store location by the gravity method.

Forecast stage: We decrease the scope of brute-force search based on gravity method and the electron clouds method. Analyzing the related data, the minimal number of the escorts is obtained by brute-force search method.

We design an algorithm for escort allocation based on time, from which the minimal total number of escorts and wheelchairs is calculated respectively. And we quantify the mutual relationships among the number of the required escort, the minimal wheelchairs and the passengers. Then we give a regression model to forecast the passenger throughput, after we can forecast the escort allocation based on time, the required escorts and wheelchairs in current period.

Implementation stage: We construct the time-split model that work with brute-force search method in practical situations. Then, we calculate the minimal required escorts according to the real-time information from the received notice. Each time we receive notice, the following steps are executed: The minimal number of the required escorts is calculated, which determines how many extra escorts are needed. Thus overtime costs treated as the primary potential costs can be estimated. If the escort number meets the requirement, the personnel allocation scheme can be determined by programming method.

Finally, we give a validity analysis of the model from the standpoint of the passengers, and then deliver our bid to the Epsilon Airline within a recommendation.

Introduction

Nowadays, air travel becomes more and more popular because it brings people great benefit. However, air travel also has some negative factors. Some survey results indicate a very high passenger's dissatisfaction rate due to frequent transitions and long waiting time at the airport. Some mandatory laws considering the handicapped people were issued up by US government, barrier-free facilities and services must be available for those help-demanded people on airports. Airlines do try to provide some special services to the passengers who are inconvenient to move, but it is always very ambiguous for an airline to devise an appropriate method to help them balance these problems (i.e., to keep costs down and to please the passengers). The outcome is always far from satisfaction to both airlines and passengers due to poor organization. Once disposed improperly, the airlines will be in severe social trouble (i.e. moral censure from the society) and suffer loss of ticket income.

Therefore, we must find the most effective method—one that not only provides satisfaction to the passengers but also allows airlines to keep their total costs as low as possible. It has great significance for the airlines to study these problems and survive in this competitive business world.

Considering these problems, our objectives are:

- to minimize the costs of the airlines, and
- to satisfy every passenger (good service will result in a good reputation and long-term profits).

In addition, our final goal is to prove the practical model to be the best one and recommend the airline to adopt our suggestions.

The first objective is the primary one. To minimize the costs of airlines, as it suggests, we should cut down any costs (e.g.: wheelchairs, rent for wheelchair storage, escort salaries, etc.). However, simply reducing the cost always leads to an extremely low profit due to the loss of the clients' loyalty. We are trying our utmost to find a perfect solution.

Here, our task comes out!

General Assumption

About the Airlines

- The airlines have enough time to schedule their working plan whenever it receives the notice (e.g.: passenger's arriving time, arriving gate, and departure gate) from the passengers who ask for help.
- Airports work all day long(i.e., 24hours), this can be achieved by simply adopting shift-working systems.

About the Escorts

- An escort can handle only one wheelchair which is available to only one passenger at each time.
- Escorts are paid annually.
- The velocity of the escorts at a the same time-slice is the same, no matter he(or she) is escorting a passenger or not.
- The velocity of the escorts who work at a different time-slice varies due to the different passenger flow.

About the Wheelchairs

- The airline should pay a certain maintenance and repair fee for each wheelchair in order to make the wheelchair in good condition (i.e., be available to use).
- The maintenance and repair fee for a wheelchair is decided by its workload (i.e., the length of the route).
- The people who ask for special help all use the wheelchairs provided by the airline.
- The time for fetching the wheelchairs is negligible.

About the Airplanes

- We ignore the weather and mechanical influence on the airplane(i.e., airplanes can arrive and departure on time).

About the Passengers

- The passenger is satisfied with the airline's service, if he is served as soon as he arrives at the airport.
- The passengers who are not specifically state in our model mainly refer to the help-demanded passengers.

Note: Our model considers the fact that the normal passengers don't need special service; this hypothesis makes it easy and quick to focus only on those help-demanded passengers.

- The passenger may never choose the same airline if he(or she) was once dissatisfied with

the service of the airline.

Note: Since it is a well-known fact that poor (or delayed) services which make passengers dissatisfied will result in loss of long-term profits. In our model, we try our best to avoid it.

About the Costs

- We don't take the costs of the other routine fee(e.g.: tax, insurance fee, etc.)into account in our model.

Note: There isn't an explicit relationship between the routine fees and other costs in our model. So we just simply ignore them. The total costs will be no apparent difference if we add constant funds annually.

Symbols

Symbol	Description
Concourse	x_{ijl} Number of wheelchairs (handled by escorts) that are sent from the initial location l to the j th arriving gate to pick up the i th-type passenger who need help.
	a_{ijl} Expense that used for wheelchair maintenance for sending the i th-type passenger from the initial location l to the j th arriving gate
	y_j Sign that shows whether passengers are waiting at the j th arriving gate
	b_{ik} Number of the i th-type passengers who are waiting at the k th departure gate
	z_{ijk} Sign that shows whether the i th-type passenger are waiting to be transferred to the k th departure gate at the j th arriving gate
	e_{ijk} Time needed to transfer the i th-type passenger from the j th arriving gate to the k th departure gate
	c_{ijk} Expense used as wheelchair maintenance for sending the i th-type passenger from the j th arriving gate to k th departure gate

d_{il} Number of wheelchairs that the location l can provide to the i th-type passengers

f_{ik} Maximum time that the i th-type passenger can use to transfer to the k th departure gate

Gate	J	Number of the arriving gates
	K	Number of the departure gates
	P	Number of the gates that the passengers are waiting at when airline receive notice
	R	Annual lease expense per square meter
Passenger	i	i represents the type of the passengers. (if $i=0$, no one sits on the wheelchair)
	G_t	Number of the total passengers at the time t
	α	Ratio of the help-demanded passenger to the total passenger
	β	Escort coefficient,(i.e., number of escorts needed per help-demanded passenger)
	T_a	Time to transfer a passenger to his (or her) departure gate
Escort	L	Total number of escorts
	T_s	Time that escorts arrive at the arriving gate
	M_v	Number of escorts work in the v th shift
	W	Annual salary of a escort
Wheelchair	E	Price of a wheelchair
	S	Average area that each wheelchair takes up
	n	Number of wheelchairs
other	C	Total costs of the airline

Note: There are also some symbols defined among our solutions, for the sake of clarity.

Approach

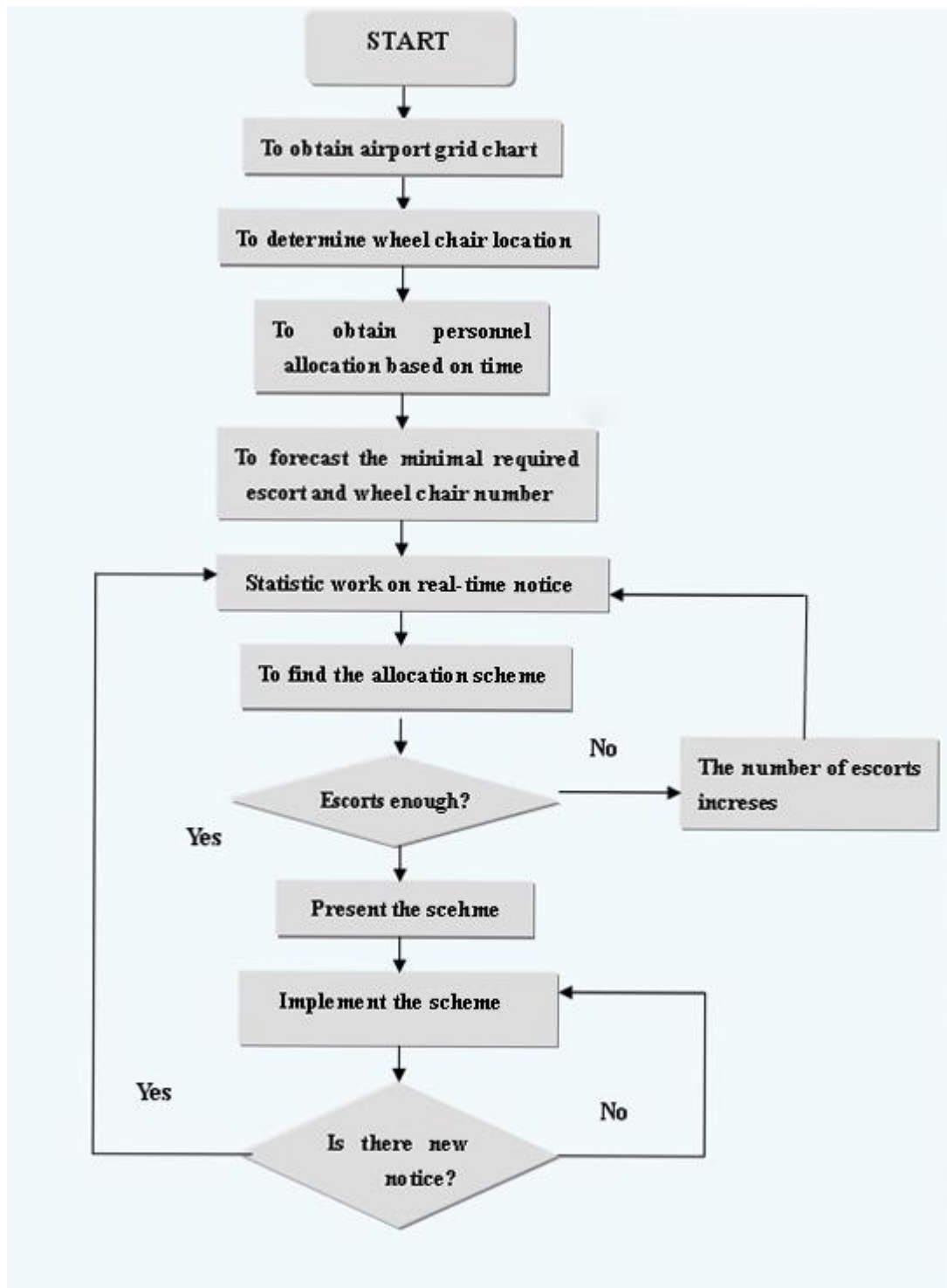


Fig.1 Ideas in our Model

Model Design

Grid Chart Model

Local Assumptions:

- We simply assume that the floor of the airport terminals is horizontal.
Note: The airport's transform from 3-D grid chart to horizontal grid chart can be achieved easily.
- We simply assume that the adjacent points are conforms equal-interval principle. (i.e., points are distributed over this grid chart equally), we define the minimal distance as d .
- We assume that all the gates are located at the points in the grid chart.

We now locate the airport terminal building in a grid chart. Applying the measurement method, we can calculate the minimal distance between any two points in this grid chart. We now use these minimal distances as 5.1 to construct a minimal distance matrix, and it is expressed as

$$U = \begin{pmatrix} a_{11} & \cdots & a_{1f} \\ \vdots & \ddots & \vdots \\ a_{f1} & \cdots & a_{ff} \end{pmatrix} \quad (5.1)$$

Where, a_{ij} represents the distance between the i th point and the j th point.

Once the distance matrix of the airport is finished, we can get the distance between any two points by simply observing the matrix.

Assume the number of the gates is N_{gate} , and the gates are denoted as g_i , where i represents the order of the gates.

Now we consider the point which stands for the i th gate in the grid chart. We can get a vector from the minimal distance matrix that represents the distances which from other points to the i th gate. (i.e., also include the points which stand for gates) It is represented as

$$\Gamma = (a_{i1}, a_{i2}, \cdots, a_{iN_{gate}}) \quad (5.2)$$

Selection of the Wheelchairs' Store Location

Since wheel chairs left in high traffic areas represent a liability risk as people try to move around them. Therefore, we should avoid those dangerous impacts caused by free wheelchairs which are stored in the terminal building. We construct a model which is suitable for location selection so that we can reduce the liable danger and make the impact probability as low as possible.

Here, we adopt the gravity method to determine the location.

The gravity method is defined as follows:

- The midpoint of all gates can be considered as vertexes.
- Connecting these vertex, we can get a polygon whose sides are not crossed each other.
- To Mark this polygon in coordinate system.

- The cg of this polygon is the location we want.
- The weight of the vertex is denoted as

$$w_i = \frac{1}{f_i}, \quad (5.3)$$

$$f_i = E(n_i) = \sum_{n=1}^{N_{\max}} n_i P(n_i) \quad (5.4)$$

Where, f_i represents the expected traffic flow of the i th gate, and

w_i represents the weight of the i th gate.

n_i represents the passenger number arriving at i th gate

$P(n_i)$ represents the probability of the n_i passengers arriving at i th gate

The reciprocal of the expected traffic flow w_i are necessary in the location selection.

Consider these phenomena:

- The **higher** traffic flow a gate has, the **farther** place we shall choose the location away from it
- The **lower** traffic flow a gate has, the **nearer** we shall choose the location away from it.

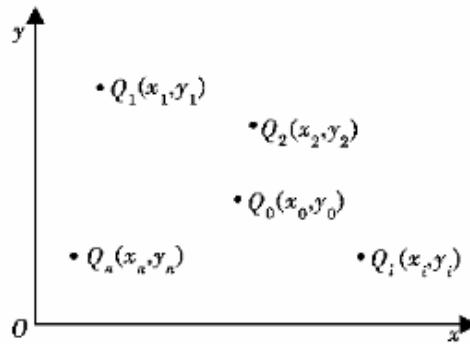


Fig.2 the Distribution of a Airport Terminal

The Distribution of an airport terminal is given in Fig. 2. From the chart we make the following observations.

- Q_i — Midpoint of the i th concourse; ($i = 1, 2, \dots, n$)
- Q_0 — Location for storing the wheelchairs;

The algorithm of the gravity methods can be presented as

$$\begin{cases} \sum_{i=1}^n w_i x_i = x_0 \sum_{i=1}^n w_i \\ \sum_{i=1}^n w_i y_i = y_0 \sum_{i=1}^n w_i \end{cases} \quad (5.5)$$

From 5.5 we obtain the coordinate of the polygon's cg $Q_0(x_0, y_0)$,

$$x_0 = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}, \quad y_0 = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} \quad (5.6)$$

Hence, we can determine the exact location of the cg (in terms of coordinates).

Applying the gravity method, we can precisely locate the position for storing wheelchairs and reduce the complexity of the algorithm. Applying this method, we can conform the security principle well and restrict the occurrence of the impact.

If we can obtain some accurate data (e.g.: locations of the concourse, traffic flow etc.), then we can determine the exact location which is used to store wheelchairs by referencing the calculations from a manual way or a computer analysis.

Brute-Force Search for Feasible Solution

We add these distances, and let the sum to be the minimal distance weight for the i th gate.

Now we consider the number of passengers who are waiting at the i th gate for help.

We combine the waiting passenger numbers and the minimal distance to define a new weight, which is denoted as $W_{i_{new}}$.

We observe the fact that:

- If the $W_{i_{new}}$ is very large, it means that a large number of people are waiting at the i th gate for help, or the average distance which from other points to the i th gate is very far which makes it uneasy for escorts to get there in very short time. Thus, we should allocate more people close to the i th gate.
- If the $W_{i_{new}}$ is very small, it means that few number of people are waiting at the i th gate for help, or the average distance from other points to the i th gate is near, which makes it easy for escorts to get there in very short time. Thus, we should allocate fewer people close to the i th gate.

Now we should devise a precise scheme to allocate escorts. Applying the gravity method, we can calculate the cg of the airport terminal building. The computing formula is given by

$$\begin{cases} \sum_{i=1}^f W_{i_{new}} x_i = x_0 & \sum_{i=1}^f W_{i_{new}} \\ \sum_{i=1}^f W_{i_{new}} y_i = y_0 & \sum_{i=1}^f W_{i_{new}} \end{cases} \quad (5.7)$$

Hence, we can determine the coordinate of the cg, which are denoted as (x_0, y_0) ,

$$x_0 = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}, \quad y_0 = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} \quad (5.8)$$

Then we can find the most likely location that is used to allocate escorts which is cg.

Here, we draw an analogy between the nucleus and the cg. Through our analytical investigation, we can obtain an exciting comparability between these two totally different notions!

We apply the basic principle of the nucleus to help us model the real issue. The similarities between them are expressed as follows:

- The probability of the location distribution of the electrons (say, escorts) is affected by the magnetism of the nucleus (say, cg).
- The electron(say, escort) rotates around the nucleus(say, cg). It appears in every location which belongs to a certain scope (say, the terminal building), and the probability of the distribution of the electrons conforms to a certain formula.

Let's assume that the two counterparts share the equivalent location distribution probability.

$$f(x) = 4 \times \frac{r^2}{a_0^3} \times e^{\frac{-2 \times r}{a_0}} \quad (5.9)$$

Where, a_0 represents Bohr Radius (5.29×10^{-11}); e equals $2.718281828 \dots$.

Applying the MATLAB program (Appendix progarm1), we simulate the electron cloud, and the fig. 3 is given below:

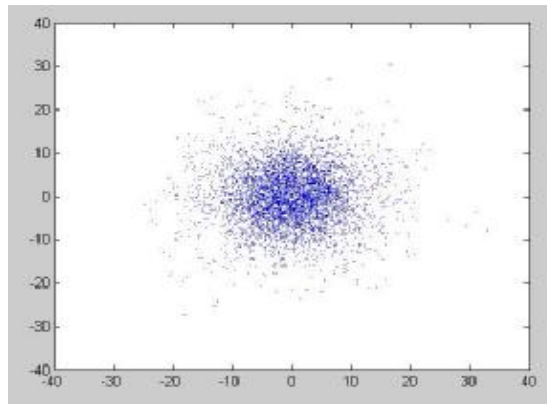


Fig. 3 Electron Clouds

During the brute-force searching, we can use the inequation given below to conforms our requirements,

$$a < n \times \int_0^r f(r) dr \quad (5.10)$$

where Δa represents the number of escorts required;

n represents the total number of escorts;

$f(r)$ represents the probability density function.

When a escorts are required, then we can use the inequation above to calculate the radius for the searching scope, this method is not only presents the feasible solution, but also cuts down the searching time.

Hence, during the process of the program that adopts brute-force to find the feasible solutions, we can first decrease the search scope by omitting those points which are far way from the cg. It leads to a high improvement of efficiency in searching process.

When we can get a search scope which is reduced, brute force search start.
We must meet some constraint condition as follows:

$$s.t. \quad g_{ij} r_{ij} < t_w \quad \forall i \in I, j \in J \quad (5.11)$$

$$e_{ijk} z_{ijk} \leq f_{ik} \quad \forall i \in I, j \in J, k \in K \quad (5.12)$$

$$\sum_{j=1}^J x_{ijl} = \sum_{j=1}^J b_{ik} z_{ijk} \quad \forall i \in I, j \in J \quad (5.13)$$

$$r_{ij}, y_j, z_{ijk} \in \{0, 1\} \quad \forall i \in I, j \in J, k \in K \quad (5.14)$$

$$x_{ijk} \geq 0 \quad \forall i \in I, j \in J, l \in L \quad (5.15)$$

Constraint condition (5.11) restricts that the time duration which is taken from the place where escort is to the Arriving GATE is less than Maximum waiting time T_w , and we just consider those escort who send 0_{th} - type passenger.

Constraint condition (5.12) restricts that the time which is needed to get to Departure Gate from Arriving GATE is less than f_{ij}

Constraint condition (5.13) ensures to balance supply and demand.

Constraint condition (5.14) restricts the values for variables r_{ij}, y_j, z_{ijk}

Constraint condition (5.15) specifies the variable x_{ijk} is nonnegative.

We begin our brute force search from the situation which the escort number is 1. If it can meet the constraint condition, and then 1 is the minimal escort number. If it can not do, we

increase one escort. Next, we continue our brute force at the situation which the escort number is 2. Just like this; we do it until get the minimal escort number which can meet all of the constraint conditions above.

Flow chart as follows:

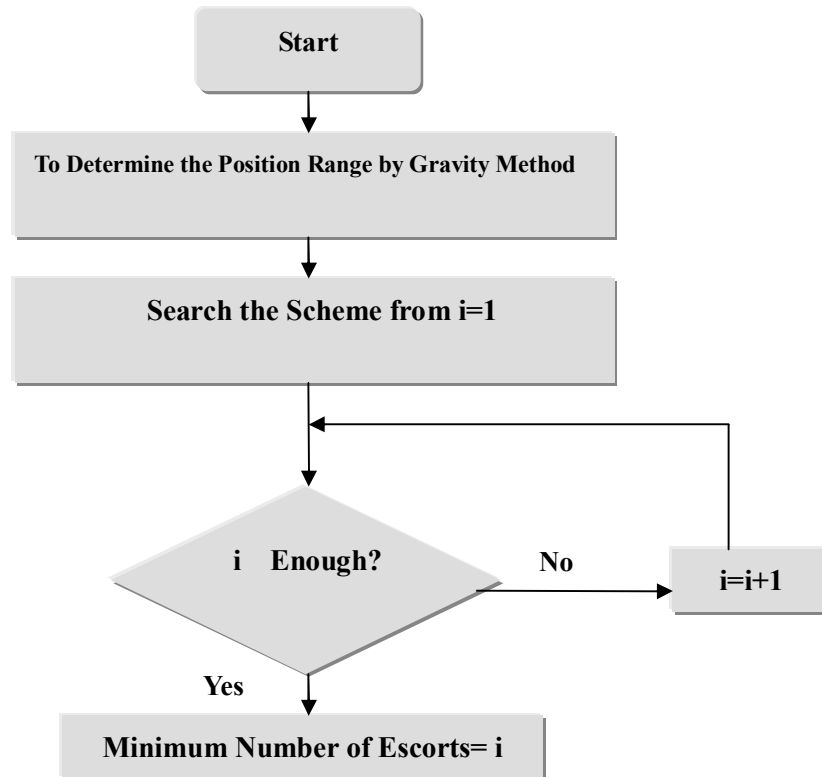


Fig.4 The ideas of the brute-force search for minimal number i

MATLAB can help us to search (Appendix program2):

The input is e_{ljk}, x_{ljk}, f_{lk} , and the $g_{lj}, r_{lj}, z_{ljk}, b_{lk}$. the value of r_{lj}, z_{ljk}, g_{lj} satisfy

Constraint condition 7 and is determined by e_{ljk}, x_{ljk}, f_{lk} .

If the result is 'mistake', it means we can't get a feasible solution, so we must increase the number of escorts to meet these constraints.

Model Refinement

There is a close relationship between the minimal distances d and the time needed to search feasible solution.

We make the following observations:

- When the minimal distance d in the grid chart is very small, It indicates that there exist immense solutions to check, and it will be impractical for a normal PC to work out the outcome.

- When the minimal distance d in the grid chart is very large, It indicates that there will be very easy to make solution check, but it will also be impractical for an airline to use this model for its failure to obtain the precision required.

Hence, the minimal distances d are one of the key issues that the airlines should pay attention to!

Since the large differences within the airports, there exist no two airports which share the same architecture. Our algorithms are depending highly on the actual architecture of the terminal building.

Here, it will be unlikely to carry out our models and algorithms and give a detailed description for the airline, because we have been isolated from the actual information about the airport.

Therefore, we are sorry to tell you that we can not present an universal model for any of the airlines. We can just give a model that highly fit a certain airlines according to its architecture and the data of the local region.

However, we can give some universal suggestions to you.

- You can make your allocation as precise as possible, and it all depends on your computer's calculation performances.
- If Pentium II (or even lower) are still used I you company, my advice is: Oh, MY God, try to purchase more high performance computers as quick as possible!!!

Calculating the value of α

α : The ratio of the number of help-demanded passengers to that of the total passengers.

In accordance to the statistics of airport passenger throughput and the number of passengers in the past years, α can be determined by data processing.

Calculating the value of β

β : The ratio of the number of escorts per help-demanded passenger.

We assume that the number of escorts is big enough. At first, only one escort work. We execute the scheme above by computer simulation to forecast the number of escorts required in different time of one day and one year. By finishing this step, β will be determined through data processing.

Total number of passengers who need help in time t is given by

$$H_t = G_t \times \alpha \quad (5.16)$$

Total number of escorts required in time t is given by

$$N_t = G_t \times \alpha \times \beta \quad (5.17)$$

Making use of computer program to figure out the statistics of H_t , the number of escorts required in different time duration in one day will be found. Personnel allocation chart based on time is obtained from the data processing.

Table 1 Personnel allocation chart based on time

Shift	Time (h)	The Number of Escorts
1	$0 \rightarrow \frac{24}{V}$	M_1
2	$\frac{24}{V} \rightarrow \frac{24}{V} \times 2$	M_2
.....		
V	$\frac{24}{V} \times (V-1) \rightarrow 24$	M_v

According to table 1, we have V shifts in one day, and the number of escorts of each shift is $M_1, M_2, \dots, M_v$ respectively. It's supposed that each escort works for 8 hours in one day and V can be divided exactly by 3 for lending itself to allocation.

General problem of linear programming can be represented as follows:

Objective Function:

$$\min z = \sum_{j=1}^n c_j \times x_j \quad (5.18)$$

$$\text{s.t.:} \quad \sum_{j=1}^n a_{ij} x_j \geq b(i=1, 2, \dots, m) \quad (5.19)$$

From the above explanation, $x_1 + x_2 \dots + x_v$ represents the number of escorts who start their work in time $0, 24/V \dots 24-24/V$ respectively. To calculate the minimal number of required escorts, we build the following model:

$$\min L = x_1 + x_2 \dots + x_v \quad (5.20)$$

$$s.t. : \begin{cases} x_{\frac{2V}{3}+2} + \cdots x_{V-1} + x_V + x_1 \geq M_1 \\ x_{\frac{2V}{3}+3} + \cdots x_V + x_1 + x_2 \geq M_2 \\ \dots\dots\dots \\ x_{\frac{2V}{3}+0} + \cdots x_{V-3} + x_{V-2} + x_{V-1} \geq M_{V-1} \\ x_{\frac{2V}{3}+1} + \cdots x_{V-2} + x_{V-1} + x_V \geq M_V \\ x_j \geq 0 (j = 1, 2, \dots V) \end{cases}, \text{ or (For } V > 3) \quad (5.21)$$

$$\begin{cases} x_1 \geq M_1 \\ x_2 \geq M_2 \\ x_3 \geq M_4 \end{cases} \quad (\text{for } V = 3)$$

With the help of software LinGo, we calculate the minimal number of escorts required in this allocation L. The number of wheel chairs equals to that of escorts in the squad which has the most escorts. Therefore, the number of wheel chairs n is calculated.

Passenger Predict Model

From the data analysis, it's common knowledge that the passenger throughput of an airport is related to the local economic development. Now we take an example to explain how to build an airport passenger throughput prediction model.

Building Regression Model

The table below is obtained by collecting the passenger throughput data from 1980 to 2001 and consulting related references.

Table 2 Related references

Year	Passenger throughput	Gross output	GDP	Per capita GDP	employee gross wages	employee mean annual money wages
1980	117411	750. 12	319. 80	541. 00	41. 63	667. 00
1981	144432	816. 27	350. 0	586. 00	44. 13	672. 00
1982	157967	897. 85	390. 17	645. 00	48. 14	703. 00
1983	127809	1014. 55	437. 65	716. 00	50. 78	723. 00
1984	192586	1239. 44	518. 85	843. 00	67. 28	931. 00
1985	242370	1642. 70	651. 82	1053. 00	86. 17	1135. 00
1986	400230	1960. 44	744. 94	1193. 00	105. 56	1327. 00
1987	533515	2472. 22	922. 33	1462. 00	121. 26	1471. 00
1988	573471	3409. 56	1208. 85	1891. 00	152. 53	1796. 00
1989	537362	3839. 67	1321. 85	2638. 00	165. 38	1918. 00
2006	MCM Honorable Mention					

1990	721594	4208. 23	1416. 50	2103. 00	184. 60	2129. 00
1991	841493	4820. 22	1601. 38	2347. 00	204. 43	2302. 00
1992	1036498	6862. 87	2136. 02	3097. 00	251. 51	2800. 00
1993	1274395	10245.27	2998. 16	4308. 00	328. 21	3615. 00
1994	1489095	14163.89	4057. 39	5785. 00	450. 44	4974. 00
1995	1952086	17699. 96	5155. 25	7299. 00	541. 62	5943. 00
1996	2181863	20804. 53	6004. 21	8447. 00	595. 72	6603. 00
1997	2111329	22314. 78	6680. 34	9344. 00	635. 4	7108. 00
1998	2094099	23433. 26	7199. 95	10021. 00	628. 49	8256. 00
1999	2166869	24578. 40	7697. 82	10665. 00	666. 28	9171. 00
2000	2436455	27241. 18	8582. 73	11773. 00	705. 36	10299. 00
2001	2793500	30074. 16	9511. 91	12922. 00	758. 55	11842. 00

With the function calculation feature of EXCEL, the correlation is figured out, which equals to 0.9969. It shows that the passenger throughput of an airport bear a close relationship with gross output in this area, GDP, per capita GDP, employee gross wages and employee mean annual money wages. These factors all meet exponential growth trend.

Further Calculation

To compute the regression model further more, the equation for passenger throughput regression model of the airport is obtained as follows:

$$Y_{\text{passenger throughput}} = 30798.8 - 87.4X_{\text{gross output}} + 1659.3X_{\text{GDP}} - 1270.5X_{\text{per capita GDP}} + 7276.9X_{\text{employee gross wages}} + 36.9X_{\text{employee mean annual money wages}} \quad (5.22)$$

Test of the Regression Model

Figure 5 shows the comparison between actual value and predicted value of passenger throughput from 1980 to 2001. Serre inequality coefficient is used for historical simulation and scale indices of ex post simulation.

In the formula, y_t^s is the analogue value, and y_t^a is the actual value, and

T is the simulative yearly number of periods.

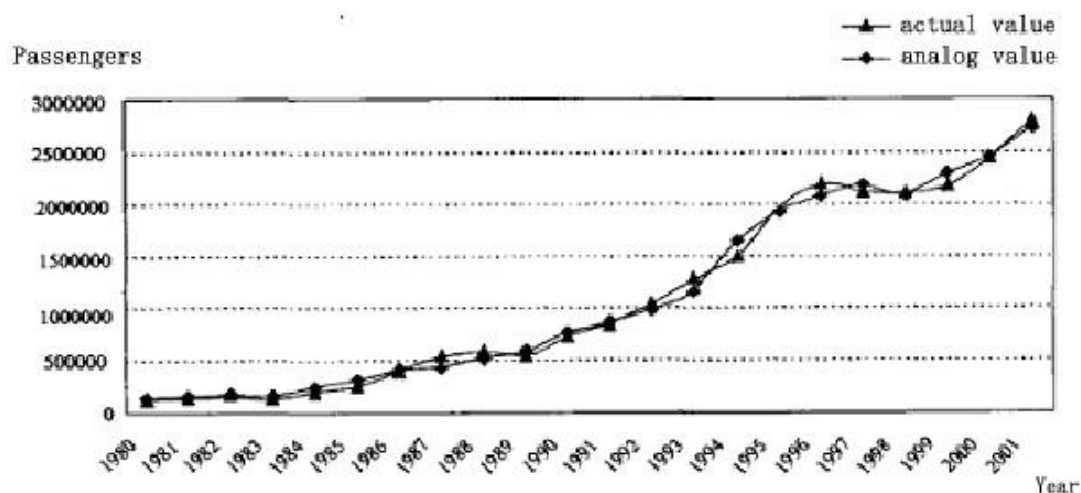


Fig. 5 Comparison between Actual Value and Analog Value

The span of Serre inequality coefficient ranges from 0 to 1.

(i) If it's 0, it means that the model fitting is very good.

(ii) If it's 1, it means that the model fitting is too bad.

Through testing the data, we know that Serre inequality coefficient is 0.0236, which means the regression model can exactly simulate the historical change trend of passenger throughput on the airport.

Determining the monthly coefficient

In order to observe the monthly variation trend of passenger throughput, it needs to make sure the percentage of monthly passenger throughput to that of last year. Then we can get observe further to the monthly value from the yearly value. Therefore, seasonal variation forecast is adopted. The linear trend without seasonal cycle is separated first. Then the seasonal index number is figured out. We use the index number to determine the percentage of monthly value to the yearly value. The monthly value acts as dependent variable while the monthly number of period acts as independent variable. Then we construct a linear trend equation. Finally, we calculate the seasonal index number of each month by moving average method from the seasonal variation trend-based forecasting.

Table 3 The raw data of passenger throughput from 1997 to 2002 on the airport

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1997	160802	152297	172795	196270	206934	163789	183313	193132	175550	199792	164064	142609
1998	158752	145503	150241	194018	189676	155689	188295	194650	173017	208685	178656	155917
1999	143291	174098	151790	184123	185867	152270	195346	222791	184421	244552	180116	148204
2000	158041	174135	163786	216974	233178	175764	235672	253597	222559	261568	183588	157593
2001	197228	140322	182822	255385	250461	204140	275611	312318	248563	286466	235609	204575
2002	195894	236647	248782	311669	292518	204978	279742	318034	278431	319024	253522	231108

After computing, the seasonal index number of every month is :

Table 4

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0.857	0.899	1.867	1.119	1.100	0.857	1.105	1.204	1.015	1.213	0.950	0.813

Divided by 12, the number of each column will be changed to the monthly coefficient

F_j :

Table 5

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
0.0715	0.0749	0.0722	0.0932	0.0917	0.0714	0.0921	0.1003	0.0846	0.1011	0.0792	0.0678

Model and Forecast

After building the regression equation of passenger throughput and determine the monthly scale factor ,the prediction model finishes the forecast of airfield handling capacity by forecasting the gross output, GDP, per capita GDP, employee gross wages, employee mean

annual money wages in the area. The five economic indicators present indices trend of development. Therefore, we adopt nonlinear smoothing prediction model. Weight coefficient a is 0. Then, we have:

(1) Yearly prediction model for airport passenger throughput

$$\begin{aligned}
 Y_{2001+i*passenger} &= 30798.8 - 87.4X_{2001+i*gross} + 1689.3X_{2001+i*GDP} \\
 &\quad - 1270.5X_{2001+i*perGDP} + 7276.9X_{2001+i*employeewages} + 36.9X_{2001+i*moneywages} \\
 X_{2001+i*gross} &= 30073.37 + 3092.02*i + 237.36*i^2 \\
 X_{2001+i*GDP} &= 9511.69 + 995.58*i + 60.69*i^2 \\
 X_{2001+i*perGDP} &= 12921.67 + 1223.41*i + 67.78*i^2 \\
 X_{2001+i*employeewages} &= 758.39 + 56.63*i + 4.82*i^2 \\
 X_{2001+i*moneywages} &= 11836.73 + 1612.31*i + 120.50*i^2
 \end{aligned} \tag{5.23}$$

(2) Monthly prediction model for airport passenger throughput.

$$Y_{(2001+i) \times j} = Y_{2001+i*passengerthroughput} F_j \tag{5.24}$$

Where F_j represents the monthly coefficient, $j=1, 2, 3, \dots, 12$, representing Jan, Feb, Mar, $\dots$ Dec respectively.

Test of the model

Table 6 the yearly predicted value of the model

Year	Forecast of gross output in past years	Forecast of GDP in past years	Forecast of Per capita GDP	Forecast of employee gross wages	Forecast of employee mean annual money wages	Forecast of Passenger throughput
2002	33402. 75	1Q567. 96	14212. 86	819. 84	13569. 53	3054857
2003	37206. 84	11745. 61	15639. 60	890. 94	15543. 33	3453859
2004	41485. 65	13044. 63	17201. 89	971. 68	17758. 12	3919617
2005	46239. 18	14465. 04	18899. 73	1062. 06	20213. 90	4452132
2006	51467. 43	16006. 82	20733. 12	1162. 09	22910. 67	5051402

Table 7 the monthly predicted value of the model and the actual value of year 2002,2003

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2002 actual value	195894	236647	248782	311169	292518	204678	279742	318014	278431	319024	253522	231108
2002 predicted value	218284	228931	220606	284848	280045	218262	281352	306511	258335	308897	241941	206981

2003 actual value	251811	236370	259826	243199	52332	134536	315089					
2003 predicted value	246794	258832	249420	322053	316622	246770	318100	346546	292077	349243	273542	234015
2004 predicted value	280075	293736	283055	365482	359319	280047	360997	393278	331464	396339	310429	265572
2004 predicted value	318126	333643	321510	415136	408136	318094	410041	446708	376496	450185	352604	301652
2004 predicted value	360945	378552	364786	471015	463072	360910	465234	506836	427173	510781	400065	342256

Table 8 the proportion of the monthly passenger throughput to the yearly one of the airport

Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
7.15%	7.49%	7.22%	9.32%	9.17%	7.14%	9.21%	10.03%	8.46%	10.11%	7.92%	6.78%

Conclusion

From the discussion above, we can draw the following conclusions:

- (1) passenger throughput of an airport have a close relationship with gross output in this area, GDP, per capita GDP, employee gross wages and employee mean annual money wages. These factors all meet exponential growth trend. The number of passengers will be 5 million in 2006 by estimating.
- (2) The year predicted value can be divided into the monthly predicted value according to the proportion showed in table 7.

Time-Split Model

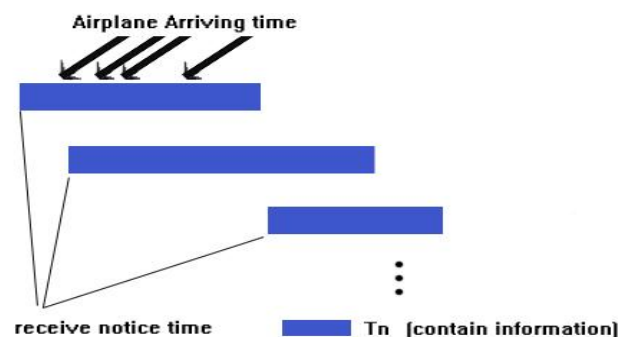


Fig. 6 Time Segment

When we receive a notice, it means we get much valuable information about following T_n time segment. The information can be used to estimate the direction that escorts flow in the future. Then the airlines can select some optimum decision point. But it is a general knowledge which the takeoff and landing of airplane is frequent, if we use the interval of two adjacent arriving time to make a decision, which is an unpractical. So we construct a simple model to split T_n into some smaller sub-segment time, which we can use to make decision and put it in practice.

Assumptions: Assuming the airplane arriving time in the following T_n time segment is

$$V_1, V_2, \dots, V_n$$

From information given by the notice, we can plot the chart and tabulate about the number of passengers who need help at each Arriving Gate. For example: figure7.

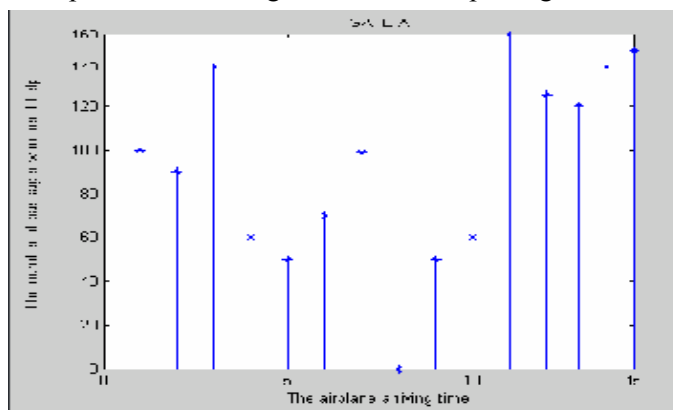


Fig.7 Variable of the Passenger Number

Definition: The variation matrix about the number of passengers who need help

$$O_{ij} = |A_{i(j+1)} - A_{ij}| \quad (i=1, 2, 3 \dots M-1, J=1, 2, 3 \dots N-1) \quad (5.25)$$

Where, O_{ij} is a matrix which has $(N-1)$ rows and $(M-1)$ columns.

A_{ij} is the number of passengers who are at GATE i when airplane is arriving.

The summation to the variation of passengers in the time interval two airplanes arrival:

$$B_j = \sum_{i=1}^{N-1} O_{ij}, \quad j=1, 2, 3 \dots M-1 \quad (5.26)$$

Time-sharing standard is defined like this:

$$B = 1/(M-1) \sum_{j=1}^{M-1} B_j. \quad (5.27)$$

It's also the mean value of total variation.

If it changes on the small side, it means the situation is stable. Therefore, the decision-making time unit can be relatively longer. On the contrary, the decision-making time unit should be shorter.

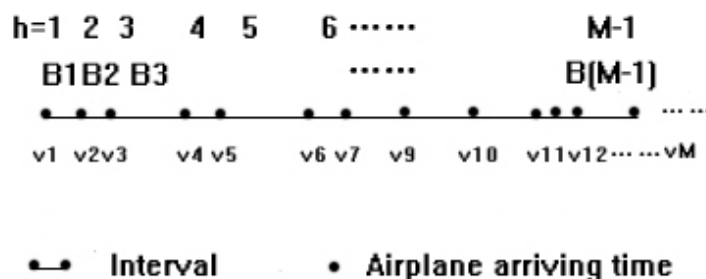


Fig. 8 Sketch Map for the Model

The cumulatively variation of passengers before the h_{th} interval is:

$$U_h = \sum_{j=1}^h B_j \quad (5.28)$$

h_{th} represents the interval of two adjacent airplane arriving time : v_h and v_{h+1}

The model need:

1 when we get a new notice, which contain new information about passengers, then we must use the time acts as a new start of an decision.

2 using function (5.28) gets U_h , if $U_h < B < U_{h+1}$, then $v_{h+1} - v_h$ is the first decision-making time unit $U_h < B < U_{h+1}$.

3 Subsequently, Judging whether there is a new notice before v_{h+1} , If the answer is affirmative, Jump to 1. Reversely, we regard v_{h+1} as a new start, and jump to step 2.

Repeat step 1 to step 3.

Escort Allocation Model

$$f = \min \sum_{l=1}^L \sum_{i=1}^I \sum_{j=1}^J a_{ijl} x_{ijl} + \sum_{l=1}^L \sum_{i=1}^I \sum_{j=1}^J c_{ijk} b_{ik} z_{ijk} \quad (5.29)$$

$$s.t. \quad \sum_{j=1}^J x_{ijl} \leq d_{il} \quad \forall i \in I, l \in L \quad (5.30)$$

$$\sum_{j=1}^J y_j = p \quad (5.31)$$

$$r_{ij}, y_j, z_{ijk} \in \{0,1\} \quad \forall i \in I, j \in J, k \in K \quad (5.32)$$

$$x_{ijk} \geq 0 \quad \forall i \in I, j \in J, l \in L \quad (5.33)$$

And we use the $s = vt$ to convert distance s to time t

For walking velocity of escort v , we consider a simple model (5.34) to bridge it and μ , μ is the number of passengers at all gate currently.

$$v = 1 / \mu \quad (5.34)$$

Constraint condition (5.30) makes the escort satisfy passengers who need help.

Constraint condition (5.31) is the number of Arriving GATE which asks for wheelchair.

Constraint condition (5.32) restricts the values for variables r_{ij}, y_j, z_{ijk}

Constraint condition (5.33) specifies the variable x_{ijk} is nonnegative.

Formula (5.29) is the objective function about the depreciation cost. It makes decision to ensure that the depreciation cost is minimal, which consists of two parts, One is the depreciation cost from the place where escort is to Arriving GATE, The other is the depreciation cost from the Arriving GATE to the Departure Gate.

We directly computer minimal expense and determine how to locate escort with Lingo or Lindo Software and by inputting the Objective function and Constraint condition.

The budget in both the short and long term

We can get an budget which can be used in both the short and long term in our model. We give this budget as follows:

- We can predict that the number of passenger will arrive in future by regression model and shift allocation model.
- We work out the minimal wheelchair depreciation expense by personnel allocation model.
- We work out wheelchair storage cost by gravity method.
- For escort payment, We know the number of escort by shift allocation model, if we must budget

the expense in future T_f years, the payment is $G_f = \gamma T_f n$. λ n represents the escort

number, λ represents the yearly salary.

To sum all above is the total expense.

Emergency Analysis

Potential expense comes from the emergency, which means the current escort number determined by the data before can't meet the effective demand of the passenger service. The required number of escorts is obtained by search method and cut-and-trial method. Overtime system makes it sure that the escort number is enough for service. It's not hard to calculate the number of escorts working overtime. The overtime cost per hour is supposed to be same for each one so that the total extra expense is determined. The model obtains the ratio of the number of passengers requiring help to the total one by Statistic in the formers. So it contains the influence of the elderly increase.

However, if the frequency of emergency is too high, it indicates that the ratio needs reappraise.

Implementation and Conclusion

Our scheme are totally worth your trust, we will give a detailed analysis report about your airline; our model covers all of the major elements of an airport simulation. Applying our model, it will lead itself to an excellent solution.

Applying the measurement method, we can calculate the minimal distance between any two points in this grid chart. Then construct a minimal distance matrix, and a weight vector for all of the gates. Further steps are supported by Step1.

Step 1 We first allocate the airport terminal building in a grid chart (given in 9).

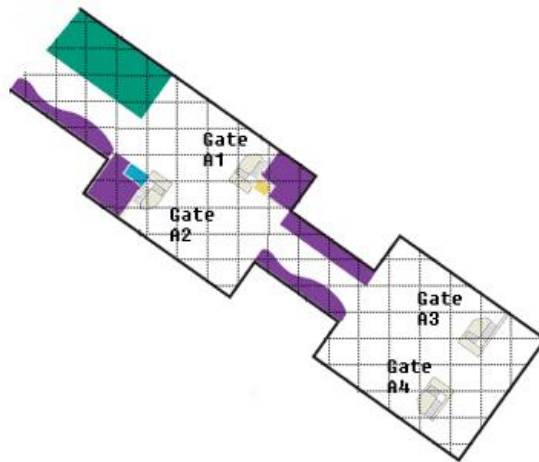


Fig. 9 Grid chart

Step2

Applying the weight of the gates which stands for traffic flow, we can obtain c_g which means the optimal place to store the wheelchairs. In that case, we can reduce the liable danger and make the impact probability as low as possible.

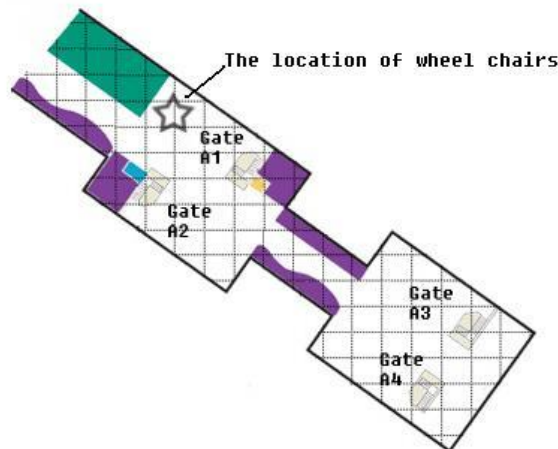


Fig. 10 Storage Location Selection

Step 3 We adjust our model to the notice given by passengers timely, then divide a whole day into several time slices. During a certain time slice, we can make an optimal allocation scheme.

Step 4

We reduce our hunting-zone and accelerate the speed of search by the notions of weight and electron clouds, then find feasible solutions by brute-force searching method, say, find the minimal number of the escorts.

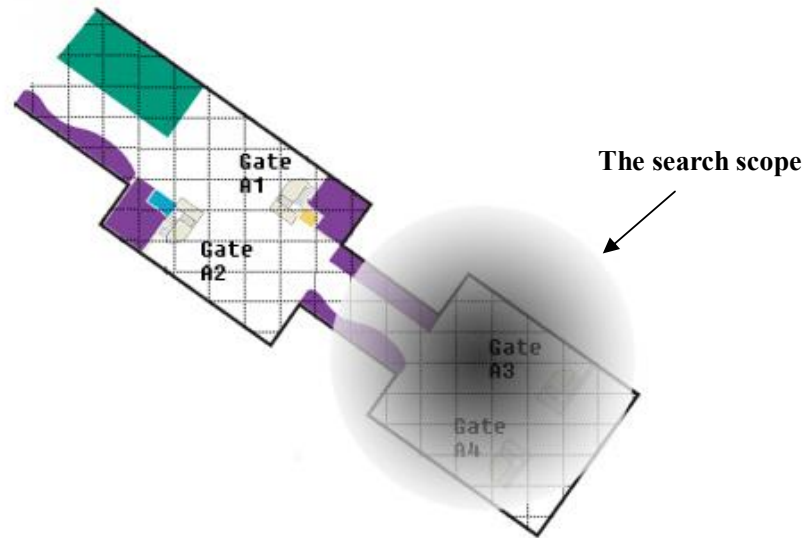


Fig.11 Search Scope Based on the Notion of Electron Clouds

Step 5

In one of the time-slices, we can obtain the optimal route-arrangement scheme according to the minimal number of escorts given by our recommendation obtained from shift-allocation scheme in the forecast stage. Be sure that the distance of the route for all of the escorts is the shortest; therefore, the expense for maintenance of the wheelchairs is also the lowest.

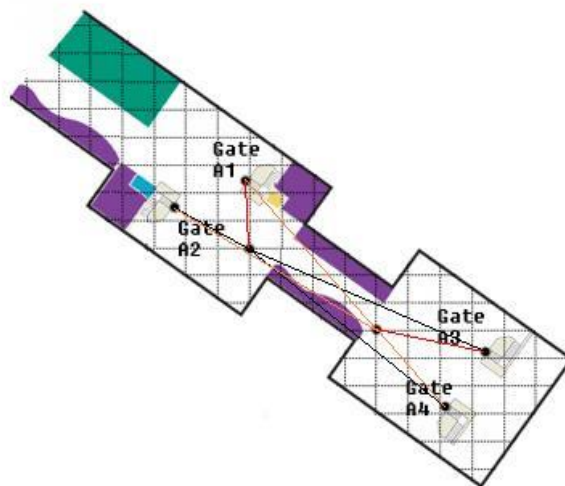


Fig. 12 Route of the Escort

Future Task

Questionnaire

In order to maximize your long-term profit, we should affirm whether the passengers are satisfied or not from their standpoints. Thus, some questionnaire surveys should be made frequently. The result of the report is really important to let you improve your service level.

Strengths and Weaknesses

Strengths

For Passenger Throughput Prediction Model

- The strengths of this model are its high practicability and flexibility. It can predict very accurate passenger throughput by year and month according to the local city data (e.g.: Gross Output, GDP, wages, etc.)
- The prediction can be refined timely by inputting the new data of the local statistic into our model. The personnel arrangement is adaptive enough to gain more profits.
- With the help of the software EXCEL, our approach shows a high efficiency in data analysis.

For Location selection Model

- The strengths of this model lies in its easy algorithm and diverse adaptation.
- Certain factors may have influence on the development of airport passenger throughout, but the inherent function of the model can adaptively adjust and draw them back to normal.

For shift arrangement model

- Applying this model, we can calculate reasonable and explicit arrangement plan with the premise of our assumptions. It meets the requirements given by the real situation well.
- Our model also takes the emergent situation(i.e., a sudden increasing number of passengers who ask for help)into consideration, and make an emergent scheme.

Weaknesses

For Passenger Throughput Prediction Model

- Our model does not take into account how different economic level affects our algorithm. This may lead to a slight deviation from the actual values.
- The model is designed for predictive analysis for 1 to 5 years. When having been used for 3 to 5 years, the model needs to be refined according to the relevant data of the local statistical data. At the same time, the monthly coefficient should be refined in the use of moving average method. With model dynamic maintenance to the model, we ensure that the predicted

value of airport passenger throughput is continuous and precise.

For Location selection Model

- The major weakness of this model is that it fails to take the actual architecture and terrain of the terminal into account. This surely comes out slight deviation, but it can at least determine a basic area for location selection.

For shift arrangement model

The major weakness is that the arrangement efficiency is not optimal enough, because we assume that escorts who work in the same day have the equivalent amount of work load (i.e., their work time is the same). What's more, we divided one day into some time segments with same length and suppose shift number can be divided exactly by 3; in fact, the efficiency is limited although it accords with reality.

Note: Indeed, We can achieve a higher efficiency if we release the constraint that we have assumed, and the realization of the perfect efficiency is based on the theoretic mathematics. However, this is not always the case. Man can't be as accurate as the mathematics can be.

Future Efforts

Questionnaire

Since the ticket incomes are the unique revenue resources for an airline. In order to attract more clients, the highest priority of an airline should be given to the problem of providing high quality services. Thus, it is necessary to establish a system which appraises the serving level from the standpoint of the clients rather than the tedious data, thereby this system can feedback information timely to the airline and let the authority of the airline make rectification timely to stop the potential drain of their clients.

From some survey results, most of the passenger's dissatisfactions are caused by long waiting time at the airport, so the average serving time can be a criterion to appraise the serving level of an airline.

Thus, we construct a $P-R$ (perception-reaction) model:

$$1 = x(t) + y(t) + z(t) \quad (5.35)$$

Here, t — Serving time (unit: s)

$x(t)$ — The percentage of people who are satisfied with the services when the serving time is t ; (%)

$y(t)$ — The percentage of people who are neither satisfied nor dissatisfied with the services when the serving time is t ; (%)

$z(t)$ — The percentage of people who are dissatisfied with the services when the serving

time is t .(%)

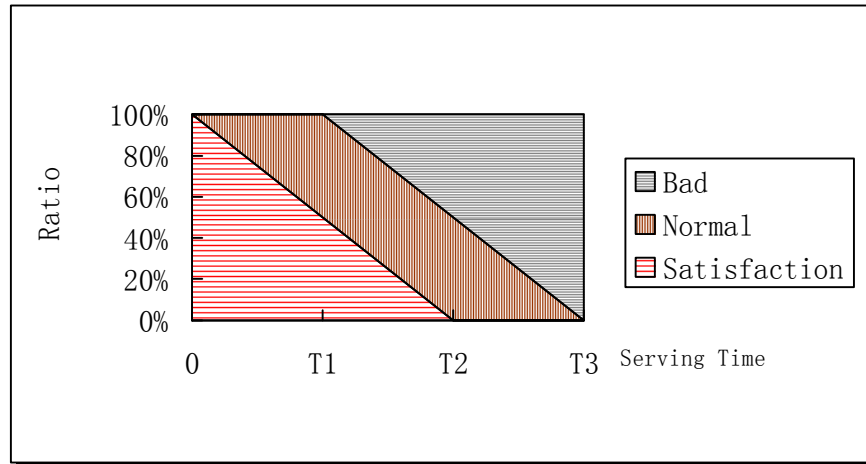


Fig.13 P-R Chart versus Serving Time

The picture of the clients' distribution is given in Fig.13 from the chart we make the following observations:

(i) For serving time T_1 , we get $x(T_1) = y(T_1)$,

T_1 represents the threshold of the **good** serving level;

(ii) For serving time T_2 , we get $y(t) = z(t)$,

T_2 represents the threshold of the **normal** serving level.

Hence, we can get the maximum time for people who can wait patiently for services. (depicted by T_1). If the average serving time of an airline are longer (e.g. the serving level are far from satisfaction) than T_1 , The airline must hasten to work out some practical methods to stop those potential loss of clients and make more long-term profits.

It is quite easy and feasible for those airlines to get feedback from their clients by simply giving out questionnaire surveys at the terminal building, then collect them and make a serving rectification according to the information which was summarized from the full and accurate analysis of the questionnaire surveys. Is there any better way than this direct way?

A Letter to the CEO of Epsilon Airline

To: CEO, Epsilon Airlines

From: MCM #170 Consultant Group

Dear Sir or Madam:

I have learned that you are planning to find a scheme to raise the efficiency of passenger service. In accordance to our allocation model, we strongly encourage you to take our advice followed which helps you get a perfect scheme:

The local relevant information, such as wage level, GDP and so on, is suggested to be collected as much as possible. It will help us forecast the passenger throughput precisely.

When having good economic returns, it's also a good idea to supply some extra barrier-free facilities besides legal requirement in order to attract more passengers. Let me take Washington International airport as an example. There are written desks used for wheelie in front of the counters, special electrical lift, restroom, etc., and it gains a good reputation from both the handicapped and the public for those negligible things.

Improving the traffic capacity on the airport can increase the transition efficiency. With the increase of the elderly, the number of wheel chairs will certainly increase, leading to the traffic jam. We suggest establishing more wheel chair access to lighten the traffic pressure. At the same time, enhancing the road capacity on the terminal and supplying passengers with convenient taxi service are also effective ways to decrease the number of passengers who are localized on the airport. For the sake of shortening the time for transition procedure, establishing special service counters is a necessary way. It helps to shorten the transition time.

The location of escorts needs to be precise and timely. In this way, we're able to obtain effective allocative decision.

In order to realize whether the passengers are satisfied, some questionnaire surveys should be made frequently. The result of the report is a part and parcel for service improvement. The shorter the passengers wait for, the more satisfied they are. Therefore, on occasions that the total expense is least, we should serve passengers as soon as possible. Concourse needs recreation facilities, to provide convenience to the passengers waiting for transition.

In a word, customers are God. It is always necessary to memorize Murphy's golden law ("If anything can go wrong, it will go")

We believe that our suggestions are useful to your decision making. We hope you will agree that contracting for our services will be of the highest benefit to your esteemed company.

Sincerely,

MCM #170 consultant group

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